

Excercise - Download climate data

Biologia della Conservazione 25/26

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- ▶ Eurostat
- ▶ NOAA
- ▶ NASA
- ▶ Our World in Data (via GitHub)
- ▶ Copernicus
- ▶ Automated Surface Observing System (ASOS)

- ▶ We will be downloading historical data from the ASOS data repository available at Iowa State.
- ▶ We will working on data from the Italian network of automated weather stations.
- ▶ First lets take a look at the website to see how we can use R to automate the process of downloading the data and eventually analyse them.
- ▶ Automated Surface Observing System (ASOS)

The packages we will be using

```
# Installing the appropriate libraries
install.packages(c("jsonlite", "RCurl", "lubridate",
                  "dplyr", "ggplot2"))
```

```
# Load libraries in the working environment
library(jsonlite)
library(RCurl)
library(lubridate)
library(dplyr)
library(ggplot2)
```

- Now that we know the web site, we can set up a script to automate data downloading

```
# first, we need to specify the date boundaries
# start date in year, month, day format
startDate <- ISOdate(1990, 1, 1)
# end date in year, month, day format
endDate <- ISOdate(2020, 12, 31)

# second, we need to specify the data network parameter
# we want to download the data from
# and the state we want to download the data from
user.network <- c("ASOS")
user.state <- c("IT") #state
```

- We can now start and retrieve the identifier of all the weather stations we are interested in.

```
network <- paste(user.state,  
                 user.network, sep = "__")  
uri <-  
  paste("https://mesonet.agron.iastate.edu/geojson/network/",  
        network, ".geojson", sep="")  
uri
```

```
[1] "https://mesonet.agron.iastate.edu/geojson/network/IT__ASO"
```

- ▶ We use the URL address we have prepared and try to retrieve the information we want.

```
# first we tell R that that we work with an URL
# URL = Uniform Resource Locator
# in this way R will be opening a connection to that URL
# and we will be able to access the website
data <- url(uri)
data
```

A connection with

```
description "https://mesonet.agron.iastate.edu/geojson/network
class       "url-libcurl"
mode        "r"
text        "text"
opened      "closed"
can read    "yes"
can write   "no"
```